

Chapter 3

Questionnaire for a description of a tonal system

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0 Generalities

The file for the Analytical Report should be in the DOCX format (or any other text format easily convertible into DOCX: RTF, DOC, ODT). The name of the file should have the following structure:

Glottolog code-language name-AR author's name

E.g.: soni1259-east_soninke-vydrin.docx

Please, indicate on which Questionnaire version your Analytical Report is based (this is version 10d).

Please ensure that tonal notation used in illustrative examples reflects all the information essential to your point (whether concerning underlying tonemes, surface tonal realizations, or both). Use several lines of representation if necessary. Whenever there is a significant difference between the underlying and the surface tonal representations, please present your example in at least two lines to make the difference clear. In such a case it is highly desirable to have at least some illustrative examples where tonal realization is indicated on every syllable or mora, in order to help readers understand the pronunciation. When relevant (e.g. when dealing with tonal spans and with extratonal segments), please use the annotation conventions, see file annotation rules.

1 General information about the language

1.1 Language name

Please indicate the name of the language, alternative names (if any). ISO-639 and Glottolog codes.

1.2 Genetic affiliation and distribution

Provide genetic affiliation of the language, its geographic area, and the number of speakers (*if relevant, especially for lingua-francas, also indicate the number of L2 speakers*).

By default, the Glottolog affiliation is taken. If you find it incorrect, please, provide an alternative genetic classification (with a reference to the source).

1.3 Dialects

List information on dialects, sociolects, etc. (especially if relevant for the tonal analysis, i.e., if tonal systems of the dialects are divergent). Please indicate which dialect(s) or variety(-ies) your AR is based on and provide the glottocode(s) (if any).

1.4 Relevant publications

Provide a concise survey of existing publications on the language, particularly on its tonal system. For well-studied languages, include only the most relevant references. The primary purpose of this section is to elucidate the present state of the study of tone in the language.

2 Segmental phonology

2.1 Phonemic inventory

2.1.1 Vowels

Please, provide a vowel chart.

Please describe the phonological status of long vowels and vowel sequences (monophonemic vs. biphonemic), if they exist in the language. The following criteria can be applied (adapted from (Culhane 2024: 49–61):

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- If [V:] and/or [V̥] have a different phonotactic distribution as compared to unambiguous sequences of two vowels (or vowel sequences are not otherwise attested), they are monophonemic (true long vowels or true diphthongs). If [V:] and/or [V̥] have the same distribution as unambiguous vowel sequences, they are biphonemic.
- If [V:] and/or [V̥] are not restricted in their distribution, they are monophonemic.
- If [V:] and/or [V̥] are in phonological or morphophonological alternation with short vowels, they are monophonemic.
- If [V̥V]/[V̥] has a vowel-like distribution, and is attested in contexts where CV/VC are not attested, it is monophonemic. If it has the same distribution as other CV/VC sequences, it is biphonemic.
- If the language has infixation, in words with a putative vowel sequence, an infix can occur between two vowels, [V:] or [V̥] is biphonemic.
- Also, languages typically have fewer diphthongs than short vowels (Chacon 2012: 30). A large inventory of putative diphthongs suggests that the language may be best analyzed as having biphonemic sequences.
- If [V:] and/or [V̥] are treated as two vowels for the purposes of morphophonological and phonological processes, they are biphonemic.
- If [V:] and/or [V̥] arise from concatenation of morphemes or other morphological processes, they are biphonemic.

Note that within the same language, some complex vocalic phenomena may pattern as true long vowels or diphthongs and others as biphonemic sequences.

It is also possible (although not common) for a biphonemic sequence to form a single vocalic nucleus, so monosyllabicity does not in itself entail monophonemicity. A complex vocalic phenomenon identified as biphonemic according to the above criteria, can be treated as tautosyllabic or heterosyllabic for the purposes of stress assignment, nasalization, or word games (Culhane 2024: 54–62).

Correlation between vowels and tone (if any).

2.1.2 Consonants

A consonant chart.

Topics which deserve special attention (if relevant):

Are there any distributive restrictions related to the consonants types? *For example, in Guro (South Mande), words with initial voiced consonants (depressors) do not appear with H tone (or do it very rarely), and words with initial voiceless consonants and sonants (anti-depressors) do not appear with L tone.*

Role of different groups of consonants in establishing boundaries of tonal spans (see Section 4.1).

2.2 Prosodic units

2.2.1 Syllable and mora

What is the minimal prosodic unit (MPU) in this language?

MPU is the smallest prosodic unit relevant for surface realization of tone. It is either syllable or mora.

By default, the MPU is considered to be the syllable, unless the contrary is proven. If the mora is relevant for the tonal system of a language, this must manifest itself somehow in the alignment of the tonal spans or tonal melodies, in the choice of allotones, variants of tonal morphemes, etc. So far, the following criteria for the relevance of mora as MPU have been formulated:

1. If there is a tonal process where the number of realized tonal targets of a morpheme depends on the number of morae in the segmental chain, mora is MPU.

E.g., in Bunoge (Dogon) the tonal marker of the head noun in an NP is L if the word contains 1 or 2 morae (CV, CV:, CVCV) or LHL if the word contains 3 or 4 morae (CV:CV, CVCVCV, CVCVCV).

2. If there is a tonal process where the position of tonal target is defined taking into account the number of morae in the segmental chain, mora is MPU.

E.g., in Aukshtaitish Lithuanian, when the locative case marker é is optionally deleted, its H tone shifts to the final mora of the preceding heavy syllable: tam-é ‘where’ → tam.

If mora is the MPU, it does not mean that other units of the prosodic hierarchy (in particular, the syllable) are not relevant for the tonal system. For example, in Iquito there is only one toneme, HLL, which is mapped onto three adjacent morae. The initial H and the final L then spread one mora to the left and to the right respectively, but only within the same syllable. E.g.,

/ariíkùmà/ ‘shoulder’ → [ariíkùmà]

/amàrijààna/ ‘year’ → [amàrijààna] (rather than *[ámàrijààna]).

Syllable structure. Discuss the problems that exist in segmentation into syllables and (if relevant) into morae.

2.2.2 Foot

Is there such a unit as prosodic foot? If it exists, what are its characteristics? Does it play any role in tonal distribution? (i.e., are there tonal rules sensitive to feet boundaries?)

It should be taken into account that a prosodic foot is not necessarily related to stress, see in particular (Culhane 2023, Vydrine 2010, Vydrin 2020).

In the languages of Southeast Asia, sesquisyllables can be regarded as equivalents of feet. In a foot, recessive syllables can display different degrees of reduction: in some languages, a full inventory of tonemes can appear on recessive syllables (although realizations of the tonemes may be of a lower intensity); in others, the inventory of tonemes on recessive syllables may be reduced (e.g., only two tonemes out of 4 or 6); elsewhere, recessive syllables may be toneless.

2.2.3 Word

Criteria for a word in the language under analysis.

In our study, we are interested primarily in prosodic word (rather than morphological, orthographic, etc.). Please explain, on what grounds a prosodic word is defined in the analyzed language.

In some languages, a word can be of little relevance (e.g., in Vietnamese, a word usually coincides with a syllable, and in Dan, with prosodic foot). For such languages, please, report cases (even marginal) where it may be still necessary to assume the existence of a word larger than a syllable/foot.

Example: in Eastern Dan, a prosodic word usually coincides with a prosodic foot. There are however several phonotypes of adjectives consisting of two feet (sometimes more) characterized by specific tonal melodies that are clear indicators of wordhood, e.g.: dēŋdēŋ ‘flat’, dīŋdīŋ ‘numb’, etc.

3 Tonal inventory

In our project, we attempt to specify the meaning of the basic terms of tonology, in particular, the term of **tone**.

We use this term

- in an abstract sense: Tone is a system of linguistic oppositions based on the pitch difference (e.g., “language X has tone”);
- in a specific sense: Tone is a tonal level or contour of yet undetermined phonological status.

This latter use of “tone” is still somewhat informal. Regarding the Jamsay Dogon form tégê: ‘s/he was/is/will be talking’, it is equally valid to say that the final mora has a L tone, that the final syllable has a HL (falling) tone, or that the whole word has a HL tone.

*To designate the acoustic reality (the F_0 curve), it is preferable to use the term **pitch** (rather than “tone”), to avoid ambiguity.*

Tonal level is a distinctive pitch range relevant for the tonal system.

Tonal target is a phonologically specified pitch target that corresponds to a tonal level. For instance, either H or L would be tonal targets.

Tonal contour is a sequence of tonal targets.

Toneme is a basic unit of tonology which can distinguish lexical and/or grammatical meanings. It may be either a level toneme (a single tonal target) or a contour toneme (a sequence of targets).¹

We analyze the aforementioned Jamsay Dogon form tégê: ‘s/he was/is/will be talking’ as containing exactly two tonemes: H spanning the first two morae and L on the final mora (Gerasimov 2024).

Tonal process is a situation where an item of mental lexicon (morpheme, stem, word, etc.) in different contexts has surface realizations that are not phonologically equivalent with respect to tone. There are two classes of tonal processes:

i) Surface mapping of tonemes onto MPUs crosses the segmental boundaries of their hosts.

ii) The toneme within the limits of an item is changed, created or deleted in a specific tonal context.

3.1 Tonal levels and contours

3.1.1 Number of tonal levels

How many levels are there in the tonal system of the language (i.e., how many tonal levels is it necessary to distinguish in order to characterize the language’s tonemes and their major allotones)?

Please, use the following universal notation:

¹In some languages, a toneme can be also characterized by other features (e.g., phonation), however, it necessarily contains a distinctive pitch (otherwise, it cannot be considered a toneme).

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Table 1: Universal notation for tonal levels

Number of levels	Symbolic representation	Tone names	Diacritics
2	H L	High, Low	á, à
3	H M L	+ Mid	á, ā, à
4	xH H L xL	+ extra-High, extra-Low; Mid	ǎ, á, à, ǎ -
5	xH H M L xL	+ Mid	ǎ, á, ā, à, ǎ
6	xxH xH H L xL xxL	+ super-High, super-Low; Mid	? -

Please, present the “traditional” notation as well (if available), and explain its correspondence to the universal notation (i.e., formulate the rule of the conversion from the “traditional” to the universal notation).

The number of tonemes (even level tonemes) may differ from that of tonal levels. For example, it is typical of privative systems of the Bantu type to have two tonal levels (L and H) and only one level toneme (most often, H, but in some languages, L). In Babanki (Wide Grassfields < Southern Bantoid < Benue-Congo < Niger-Congo), there are 4 tonal levels and only 2 level tonemes (H, L): the M tone only arises as a realization of the combination /HL/ hosted by one syllable, while xL is only attested phrase-finally as part of the LxL realization of /L/ (Akumbu & Vydrin 2024). Mandarin has 4 tonemes (one level and 3 contour tonemes) that are described through reference to 4 or 5 tonal levels (depending on the dialect and the author).

3.1.2 Contours hosted by one MPU

Arguments for and against their tonemic status are discussed in Section 3.2; please don’t discuss this question here.

For designation of simple tonal contours, please use combinations of the letters suggested in Section 3.1.1, and simple diacritics: circumflex (â) for the falling contour, HL, and hachek (ǎ) for the rising tonal contour, LH.

For richer contour systems, please, use combinations of numbers from 1 to 5, where 1 stands for the lowest level, and 5, for the highest. For example, in Wobe (Kru < Niger-Congo): pa⁴¹ ‘cocoon’, pa⁴² ‘to carve (intr.)’, pa⁴³ ‘to carve (tr.)’, pa³² ‘to enter’ (Bearth & Link 1980).

3.2 Inventory of tonemes

Please, enumerate all tonemes of the language, and mention their regular allo-tones.

For each toneme, enumerate the criteria of tonemic status it complies with. For each case, please either provide diagnostic examples or make references to subsequent sections (e.g., Section 6.1 on tonal processes), where relevant data is discussed in more detail.

There are three types of criteria for toneme identification.

1) A general criterion.

- **The Persistence Criterion:** If a tone of a morpheme that is realized as a single MPU is contrastive and it persists in all contexts (i.e., it is not affected by tonal processes which could be regarded as criteria of tonemic status), it is considered to be a toneme.

This criterion is crucial for those languages where tonal processes are lacking or marginal. It is applicable both to level and contour tonemes.

To avoid populating tonal inventories with only marginally attested tonal contours, contour tonemes identified on the basis the Persistence criterion must also comply with the contour entrenchment condition:

If a tonal contour is lexically specified for a prosodically minimal morpheme (one MPU), this tonal contour is a toneme if:

- a. it is frequent in texts, and / or
- b. it is productive (i.e. it is frequent in the dictionary or in inflectional paradigms).

It is hardly possible and maybe not even desirable to set a fixed frequency threshold to apply the contour entrenchment condition. It is left to the discretion of the author of the analytical report.

2) Non-zeroneess criteria

These criteria concern level tones and are meant to answer the question: does a tone have a tonemic status or is it phonologically a zero? The essential idea on which all our non-zeroneess criteria are based is that default (zero) pitch is not informative. If some tonal target is realized in a situation of articulatory or time constraints, it is informative and hence not a zero.

Similarly, if a tonal target is preserved when displaced relative to the segmental boundaries of a morpheme to whose lexical specification it belongs, it is likewise informative and non-zero.

The following criteria of the tonemic status are applied:

- **The Shared MPU Criterion:** If two level tones can be assigned to one MPU, each tone is a toneme or a part of a toneme, and neither is $\emptyset$. *This criterion is applicable to tonal contours whose status of single tonemes has not been confirmed along the criteria for the contour tonemes, see below.*
- **The Activity Criterion:**² If the presence of a tone correlates with the presence of a morpheme and this tone is able to surface or trigger a tonal change outside the segmental exponent of this morpheme,³ it is a toneme or a part of a toneme. *This criterion can be represented in a simplified formulation: only tonemes, but not zero tone, can trigger tonal processes. As there are many types of tonal processes attested, there are many ways in which a tone's active character can be manifested. This criterion thus covers a wide range of cases: tones that are spread or copied across morphemic boundaries, tones that trigger polarization or other changes in adjacent tones, floating tones (Section 3.3; in the vast majority of cases, a floating tone is postulated on the basis of some tonal process), grammatical tones (Section 7.1; by definition, these surface outside the segmental exponent of their morpheme, if any.), etc. When you identify a toneme based on the Activity criterion, please try to list all relevant contexts (if only with brief references to other sections), so that relative activity of each toneme could be evaluated.*
- **The Intonational criterion:** If a tonal target is maintained in an interaction with an intonational pitch/contour, either at the expense of the non-realization or modification of the intonational tone, or by a shift of a lexical tonal target to a different segmental position, this tonal target is a toneme or a part of a toneme. *In other words, if a tone is different from what can otherwise be shown to be the default pitch*

²The Activity criterion was first formulated by Hyman (2000), whose generalization (“if the opposition is /H, $\emptyset$ /, tone rules should refer only to H’s”) we took as a departure point, and further elaborated. Another proposal discussed in the same paper (‘only specified tones can float’) is also subsumed under the current formulation of the Activity criterion.

³This wording is intended to also cover tonal morphemes: as they lack an overt segmental exponent, their tonal exponent is necessarily realized on segments that belong to other morphemes.

pattern on extratonal MPUs, it is a toneme or a part of a toneme. E.g., in Poko-Rawo (McPherson & Dryer 2021: 12) there are two classes of prosodic words that generally appear M-toned in non-phrase-final position and are targeted by H tone spreading from a preceding prosodic word. In phrase-final position in the absence of tone spreading, however, words of the first class also appear with a level M tone, while those of the second class display a falling pitch. Words of the first class can be analyzed as carrying a M toneme (which cannot be identified on the basis of any other criteria); the second class comprises inherently toneless words that receive a default pitch (falling phrase-finally, M otherwise). In other words, the M toneme in the language is not phonologically “active” enough to trigger tonal processes, but “active” enough to block the boundary tone.

If a tonal system in question has two tonal levels, and only one can be recognized as a toneme, and the other one is a zero tone (i.e., a tone which is not a toneme), such a system is privative, i.e. (H, Ø) or (L, Ø). If both tones are tonemes, it is omnitonal (L, H; the term “omnitonal” was recently coined by Larry Hyman, p.c.). A system with three or four level tones can also be interpreted as including two tonemes and a Ø tone (H, L, Ø) (Kugama seems to have such a system).

3) **Contour criteria** are designed to distinguish between contour tonemes and combinations of level tonemes.

- **The Non-Compositionality Criterion:** If a language has a tonal contour realized on one MPU composed of levels at least one of which is not available in this language as a toneme, this contour is a toneme.
- **The Holistic criterion:** If a tonal specification of a morpheme consists of a sequence of tonal targets, and none of the targets can be shown to constitute a toneme in the language, this sequence is a toneme.

The difference between the two contour criteria (which are otherwise both based on the same principle) is that the holistic criterion can be applied to segmental units that are not prosodically minimal.

For contour tonemes that can be extended over multiple MPUs, please describe how tone breaks are localized. Are their positions determined by general mapping rules or lexically specified? For instance, in Japanese (which has one toneme, /HL/), each tone-bearing morpheme has an **alignment point** at a specific moraic boundary that corresponds to the juncture

of *H* and *L*, while tonal span boundaries are established by the rule of tonal spreading (Polyakova 2026).

The criteria above are based on the most general theoretical expectations about the toneme as an item of the phonological inventory and about its relation to the segmental chain. We postulate the following **toneme properties**:

- Scalability: *A toneme can be extended to multiple consecutive MPUs. Two consecutive tonemes can be compressed onto one MPU.*
- Non-obligatoriness: *Not every MPU is necessarily associated with a toneme.*
- Continuity: *A toneme is realized on an uninterrupted sequence of MPUs.*

3.3 Floating tones

A **floating tone** is a tone that is present in a phonological representation (underlying or intermediate), but is not associated to any segmental unit. In the course of the derivation, a floating tone either gets associated (“docked”), realized indirectly (e.g., as a downstep of the following tone), or deleted.

The notion of a floating tone is a powerful analytical tool that has proven its usefulness in description of diverse tonal languages and has greatly influenced the development of autosegmental phonology. At the same time, the very presence of floating tones in a fragment of data is dependent on a specific analysis; there also may be subtle but important differences in the actual application of the term by different authors. For this reason, we do not aim to put forward any typological generalizations in terms of floating tones.

We still include this section for the sake of completeness of description of particular languages. It remains up to the authors when and where to postulate floating tones, but we caution against over-reliance on this analytical tool. We suggest the following guidelines

- (a) Tones that are only “floating” at the underlying level, but get associated to segments of their host morphemes according to general mapping rules of the language are not worth mentioning in this section (in an autosegmental model with no lexical pre-association, all tones are underlyingly “floating”, which renders the term too broad for any practical usefulness);
- (b) Grammatical morphemes that have tonal, but zero segmental specification are covered in Section 7.1; it makes sense to mention them here only if they show some variability of realization.

Are there floating tones in your language? To which level of representation can they be attributed? (*the basic underlying level? an intermediate level, where a floating tone can result from tonal processes, such as toneme delinking?*)

Realizations of floating tones. What are the rules of their association to segments at the surface level?

Functions of floating tones. Do they represent separate tonemes or constitutive elements of tonemes? Do they represent tonal morphemes?

3.4 Downstep and upstep

We define **downstep** as any downtrend (lowering of the phonetic backdrop against which the phonologically specified pitch targets are scaled) caused by a local trigger. The latter requirement sets it apart from declination, which is a global, utterance-level downtrend covered in Section 6.3. Downstep can be triggered by an adjacent tonal target, an adjacent floating tone, a prosodic boundary, etc., or even be an independent feature in a tonal specification of some morphemes, see (Lionnet 2022) on Paicî, (Rochant 2023) on Baga Pukur.

Historically, there has been much terminological variation in the discussion of phenomena subsumed under the label of “downstep” here. There is a tradition of calling a downtrend triggered by a preceding lower tone “downdrift”, reserving the term “downstep” only for cases without any visible trigger; Stewart (Stewart 1965) proposed “automatic” vs. “non-automatic downstep” for the same distinction. We find both these solutions unsatisfactory for a number of reasons. For a typological study, it makes sense to define downstep broadly but pay attention to the parameters of variation.

While downstep is local with respect to its trigger, it is global in its domain, as all subsequent tones until the end of the utterance (or the phonological phrase) are affected (Connell 2002: 4). There are, however, cases when a single tonal target gets realized lower, with no effect upon the following tones, as in Dom (Gerasimov 2025) or Mambila (Connell 2002). Such cases pertain to the discussion of allotony in Section 3.2.

Are there downstep phenomena in the language? What are their targets and triggers?

Can the nature of downstep be characterized as assimilatory, dissimilatory, or neither?

Is downstep phonologically contrastive? Can downstep provide evidence for the presence of a toneme or a tonal process that may not be identifiable otherwise?

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How significant is the lowering of the phonetic backdrop? Is downstep total or partial?

In some languages, an affected tonal target is lowered all the way down to the next contrastive pitch level, so that a H^LH sequence is indistinguishable from HL (hence, “total downstep”). More often, however, lowering is not so drastic. See (Stewart 1993) for a discussion.

Is there upstep in the language? If so, please answer the same question about it.

***Upstep** is defined as the mirror image of downstep: upward shifting of the phonetic backdrop against which the phonologically specified pitch targets are scaled caused by a local trigger. It appears to be a rather infrequent phenomenon in the world languages (Hyman & Leben 2021: 51).*

3.5 Other suprasegmental features of tonemes, apart from pitch.

Interaction of tonemes and phonations. Registers

Phonation and other laryngeal features. If the language has these features, can they be regarded as integral parts of tonemes, or are they independent?

Does your language have registers?

Registers (a term widely used in the study of Southeast Asian languages) are more or less abstract notion, it may have variable manifestations: pitch, laryngeal features, vowel quality or quantity; for a survey see (Kirby & Brunelle 2017). If you language has registers, please describe their phonetic manifestations, especially with respect to the interaction with pitch.

The term “register” is sometimes also used in a different meaning, as a basic pitch level in relation to downstep and similar processes (Snider 2020). In this section, we use the term “register” in the SE Asian sense.

4 Tonotactics: tonal span, tonal phrase

4.1 Tonal span boundaries

Tonal span is the part of the segmental chain on which a toneme is realized on the surface level.

4.1.1 Tonal span size

What can be the size of a tonal span (in syllables or morae)? Please, indicate its minimal and maximal size (if relevant).

The size of a tonal span can be $\emptyset$, if, for example, a floating low tone is realized as a downstep. If two tonemes are realized on one MPU (i.e., if we have a contour which is interpreted as a combination of two level tonemes), the size of a tonal span is $\frac{1}{2}$. In some languages, the upper size of a tonal span is theoretically unlimited.

4.1.2 Establishment of tonal span boundaries

Are there tonal rules that affect the boundaries of tonal spans? Please, mention such tonal rules, if available (with reference to Section 6.1).

4.1.3 Tonal spans and prosodic units

Two kinds of correspondence between the tonal span and prosodic units can be imagined:

- a) Coincidence: tonal span limits obligatorily coincide with the limits of a prosodic unit (e.g., in Vietnamese, a tonal span always equals a syllable).
- b) Conformity: a tonal span tends to occupy the entire prosodic unit of a certain level (cf. the conformity as a toneme property mentioned in Section 3.2). *The conformity can manifest itself through tonal rules (e.g., there may be a tonal spreading rule resulting in the extension of the limits of a tonal span to entire foot or word, etc.), or in the mapping of grammatical tones (e.g., in Eastern Dan, replacive tonal morphemes are always mapped onto the entire foot).*

Please, mention if there is any kind of correspondence between tonal spans and prosodic units (mora, syllable, prosodic foot, prosodic word, prosodic phrase)? If such a correspondence exist, specify which type of correlation is available (coincidence or conformity). For conformity, please, specify how exactly it manifests itself.

Less probably, there might be a correspondence between other types of units: morpheme, root, stem, morphological word. If such a correspondence is available in your language, please, characterize it.

4.2 Combinations of tonemes. Tonal melodies

If boundaries between tonal spans and word/prosodic foot boundaries do not coincide: are there restrictions on combinations of tonemes within a word/foot?

Is the notion of tonal melody relevant for the language in question?

In our terminology, a **tonal melody** can be preliminarily defined as a stable combination of tonemes mapped on a segment (most often, a word) of a variable size. The notion of tonal melody is close to that of a restriction on combination of tonemes in a word.

4.3 Extratonal syllables

Are there extratonal syllables (i.e. syllables not included into tonal spans)?

In some theoretical models, if a tonal span exceeds one MPU, all its MPUs but one can be considered as underlyingly toneless; but this is not what is meant here. Extratonal syllables are those which do not belong to any tonal span on the surface level.

If yes, how are they tonalized at the surface level? Options:

- they obtain a default tone (which is not a toneme),
- they are tonalized through polarization, and the tones generated through polarization are not tonemic;
- other?

4.4 Tonal phrases

A tonal phrase is a maximal domain for tonal processes consisting of two or more prosodic words.⁴ A tonal phrase in an individual language should be defined in syntactical terms. *Are there tonal phrases in your language? If they exist, which syntactic constructions represent tonal phrases? Which tonal processes occur in the tonal phrases?*

5 Stress and tone; culminativity; prominence; obligatoriness

(We avoid the term “accent” which is vague and variably defined by different authors; seemingly, the realities usually represented in the terms of “accent” can be more precisely represented through other notions.)

⁴Cf. the definition of a phonological phrase by Chomsky & Halle (1968: 9) as “a maximal domain for phonological processes”.

5.1 Culminativity

Is tone in the language in question culminative? *Culminativity means that there can be, at maximum, only one toneme (underlying tone) per word. Culminativity does not require every word to have a toneme; a system can be regarded as culminative even if there are toneless words.*

5.2 Stress

Does the language have stress? *(Stress is obligatory, it must be present in every prosodic word; it is also culminative: a word can have only one (main) stress. Stress is necessarily associated with an entire syllable, rather than a mora.)* If the language has stress:

5.2.1 How is stress expressed? (pitch; vowel length; vowel inventory, consonant length and/or inventory, etc.)

5.2.2 Is position of stress fixed or free?

5.2.3 If it is fixed, what is its position?

5.2.4 If position of stress is free, does it change in inflexional paradigm? through derivation?

5.2.5 Does the language have secondary stress?

5.2.6 Is there any correlation between stress and tonal distribution? (e.g.: tonal contrasts are manifested only on stressed syllables; position of the tonal span is predictable with relation to the position of the stress; tone and stress do not correlate; other?).

5.2.7 For pitch-stress languages: are there stressed syllables on which the tonal contrast is not manifested?

For the pitch-stress languages, the accentuation rules are usually formulated in terms of “recessive” and “dominant” (“strong”, “accent-attracting”, etc.) syllables. Please, convert this into the format of tonal rules. E.g., in Aukshtaitish Lithuanian, according to the “accentual” interpretation, syllables are subdivided into dominant and recessive. If a word contains a dominant syllable, this syllable is stressed (and if there are two dominant syllables, the first one is stressed). If there is no dominant syllable in a word, the first recessive syllable is stressed. According to our interpretation, dominant syllables are underlyingly tonal (i.e., tonemes are associated with them). Thus if a dominant

syllable surfaces as toneless, this must result from a toneme deletion rule. If a recessive (underlyingly toneless) syllable surfaces as tonal, we must postulate a toneme creation rule.

5.3 Positional prominence

(For languages that have no stress:) Is there any kind of positional prominence? *Stress is also a kind of syllabic prominence which is obligatory. Here, we speak of syllabic prominence which is not obligatory (i.e., a prosodic word can have no prominent syllable).*

5.4 Obligatoriness of tone

(For the languages which have no stress:) Is this language characterized by obligatoriness of tone? *The obligatoriness means that any prosodic word either bears at least one toneme or is included into a tonal span of a neighbouring word.*

6 Tonal rules. Segmental rules which have impact on tones

6.1 Tonal rules (tonal processes)

List all the tonal rules relevant for your language (normally, rules explain the passage from underlying tonemes to their surface realizations) and describe their sequence (for each rule, indicate which other rule it follows) and constraints on their application.

For each rule, indicate if it affects the tonal density (through erasing underlying tonemes, etc.).

Tonal rules:

- tonal spreading,
- tonal shift,
- toneme copying,
- toneme deletion,
- fusion of tonemes,
- plateauing,
- Obligatory Contour Principle (polarization),
- toneme creation.

If application of a tonal rule is conditioned by morphosyntactic factors (part of speech to which a word belongs, etc.), please, mention this fact.

6.2 Segmental processes affecting mapping of tonal spans

List the rules concerning segmental units which affect the mapping of tonal spans (e.g., a vowel elision resulting in resyllabification, thus increasing or decreasing the number of syllables).

6.3 Tone and intonation

If there are cases where intonation influences tonal realizations (or vice versa), they should be presented here. *Since tone and intonation utilize the same medium of pitch, situations may arise when the two compete for realization. Intonational phenomena, such as peaks, boundary tones, etc. can block tonemes from realization or displace boundaries of their tonal spans; or they can be similarly affected by tonemes.*

Suspension of declination or downstep for intonational purposes can also be mentioned here.

Declination is an automatic gradual downward modification (over the course of a phrase or utterance) of the phonetic backdrop against which the phonologically specified pitch targets are scaled. Unlike downstep, declination cannot be ascribed to any local trigger corresponding to a specific morpheme or a specific position in a segmental chain

It is typical for human languages to suspend declination in non-declarative utterances. However, downstep can also be suspended in a similar manner. Thus in Hausa questions, as shown in Lindau (1984), Inkelas & Leben (1990), all H tones are realized at approximately the same level (with a slight declination), regardless of any intervening L tones, which would trigger downsteps in declaratives.

7 Grammatical and lexical tones

7.1 List of grammatical tonal morphemes

List all morphemes in the language under study that have grammatical tones in their lexical specification.

*Grammatical tones are tonemes, combinations of tonemes, or tonological operations that serve to distinguish grammatical meanings.*⁵ Two types of morphemes

⁵For the purposes of this project, we (rather boldly) assume that all morphemes in a language can be unambiguously divided into lexical and grammatical. Any problematic cases have to be decided on a case-by-case basis. The inflection vs. derivation distinction is of little relevance for our purposes and the label “grammatical morphemes” subsumes both inflectional and derivational morphemes.

can supply grammatical tones:

- *tonal morphemes: grammatical (inflectional or derivational) morphemes that have only tonal, but no segmental exponent;*
- *grammatical morphemes that have both segmental and tonal co-exponents,⁶ but whose tonal exponents are realized (a) on segments outside the boundaries of their segmental exponents; and (b) in a manner specific to this morpheme (or a natural class of morphemes that includes it), rather than according to the general phonological grammar of the language⁷.*

For each morpheme, specify its meaning(s), segmental (if any) and tonal exponent, position of the former and details of realization of the latter, allomorphy, etc.

A tonal exponent of a morpheme can be a sequence of tonemes specified in phonological representation or an operation performed upon tonemes supplied by other morphemes. It is thus possible to speak about **item-based** and **process-based** grammatical tones, following (Sande 2022: 401–403).

Item-based grammatical tones are those that can be described within the framework of additive morphology, i.e., where the meaning can be mapped onto an underlying form (or a set of alternating underlying forms in case of **suppletive grammatical tone**).

The item-based grammatical tone can be further subclassified by several parameters:

- **replacive grammatical tone** which deletes the lexical tones of a word (or of a segment of a word) and **additive grammatical tone** which is docked on a segment without deleting its lexical tone;

⁶Note that such cases (when a tone and a segment participate in exponence of one and the same morpheme) are different from true multiple exponence where both exponents are attested independently of each other and can therefore be viewed as separate morphemes (Harris 2017: 17–21), see (Kaldhol 2024) for discussion.

⁷There are subtle differences between our understanding of grammatical tone and the definition proposed in Rolle (2018: 19): “a tonological operation which is not general across the phonological grammar, and is restricted to the context of a specific morpheme or construction, or a natural class of morphemes or constructions (i.e. grammatically conditioned tone addition, deletion, replacement, shifting, assimilation, dissimilation, etc.)”. For instance, in the Jamsay Dogon form *tégé*: ‘s/he was/is/will be talking’ (from *tégé*- ‘talk’) we identify the L toneme on the final mora as the grammatical tone, while under Rolle’s definition the term applies rather to the **operation** of this toneme’s addition. We still believe that the same set of natural language phenomena would call for postulation of grammatical tones under the two approaches.

- **dominant grammatical tone** which deletes lexical tones of the word, and **recessive grammatical tone** which is docked onto originally toneless segments of a word.

The **process-based grammatical tones** are those which cannot be reasonably derived from an underlying representation, but can be represented as processes. Process-based grammatical tones include phenomena such as:

- **mirror-reflecting grammatical tone**: in a 2-level tonal system, a grammatical meaning is expressed by replacement of each toneme by the opposite one, like the Nominative vs. Oblique cases marking in Kipsigis (Sande 2022: 415–418);
- **subtractive grammatical tone**: a meaningful deletion of the lexical tone of a morpheme without replacing it by another toneme, thus leaving it toneless;
- **scalar shift**: a grammatical meaning is expressed by rising or lowering of the tone by one or two degrees. So, in a 4-level tone language Guébie (Kru), the imperfective is expressed on the verb through lowering of tone in one step ($xH \rightarrow H$, $H \rightarrow L$, $L \rightarrow xL$); and if the lexical tone of the verb is xL , then the tone of the preceding word (the subject) rises in one step (Sande 2022: 419–420); cf. also a more complicated subtype represented by the conjoint verbal construction in Eastern Dan (Vydrin 2024).

Of relevance for the understanding of grammatical tone is the notion of realization window⁸ which refers to a segment of the target word on which the grammatical tone is realized. A realization window can coincide with a prosodic unit (a mora, a syllable, a foot, a prosodic word, a tonal phrase), or its position can be defined by a more or less complex set of rules. E.g., in Mwan, the realization window of the Imperative/Optative grammatical tone morpheme (a M tone) is the second syllable of the verbal stem, if the stem is represented by a heavy foot; this grammatical tone is not expressed on a stem represented by a light foot (Perekhvalskaya & Vydrin 2024).

There is an important convention to be taken into account in the annotation of texts: there are cases where the result of application of a grammatical tone to a morpheme is identical to the latter's lexical tone, so it is not immediately clear which one is realized. In the absence of language-specific arguments in favor of

⁸Our “realization window” is equivalent to Rolle's (Rolle 2018: 29) “valuation window”.

one or the other solution, we generally assume that if it is otherwise proven that the grammatical tone in question is dominant, it is the grammatical tone that finds realization, as this results in a simpler and more regular system.

Grammatical tone is not to be confounded with the cases where a toneme is assigned underlyingly to a segmental affix and is mapped on this affix superficially; what we have here is a mere lexical tone of the affix.

7.2 Tonal paradigms

If there are tonal grammatical paradigms, describe these (however, if such paradigms are very bulky, their detailed presentation may be superfluous for the understanding of the functioning of the tonal system and calculation of TDI; in such case, a concise presentation may suffice).

7.3 Lexical tones

Does language have lexical tones? *The great majority of tonal languages have lexical tonemes (and often enough, only lexical tones are available). However, there are also languages where lexemes have no lexical tonemes; they are tonalized by grammatical tones or through tonal processes.*

If a language has lexical tonemes, but there are words without these, please specify what kind of lexemes are toneless:

- grammatical classes of lexemes;
- words of similar origin (borrowings, lexifiers);
- they are randomly distributed?

If there are no toneless words in the language, please, say it explicitly.

7.4 Tonal specification of bound morphemes

7.4.1 Toneless bound morphemes

Are there underlyingly toneless morphemes? (i.e., morphemes to which no toneme is attributed at the underlying level)

If there are, how are they tonalized superficially? *Whenever necessary, refer to the appropriate tonal rules described in Section 6.1.*

7.4.2 Tonally specified bound morphemes

Are there tonally specified bound morphemes? If there are, explain how their tonal interaction with the stems.

8 Tonal classes of words

8.1 Differentiation of parts of speech by tone

Are there any differences in the tonal behavior of words belonging to different parts of speech? E.g. some tonal combinations are allowed only on verbs or on adjectives; some word classes may have no lexical tones (verbs in certain African languages), etc.

8.2 Tonal classes of words (not necessarily related to the part-of-speech attribution)

In some languages words are divided into classes (which may correlate with their division into parts of speech, but not necessarily), and members of these classes differ in their tonal behavior (in a way that this difference cannot be explained through floating tones). For example, in Mwan (< South Mande < Mande < Niger-Congo), there are two tonal classes: “constant” (words with unmodifiable tones) and “mobile” (words with tones sensitive to grammatical context), see (Perekhvalskaya & Vydrin 2024). Are there tonal word classes in your language? If there are, describe their particularities.

9 Diachrony of tone

Provide any data on the diachronic processes in the tonal system if available. (For languages without long written tradition, as a rule, such data are absent. For such languages, if there is any attempt of tonal reconstruction, its results can be given here.)

10 Traditional representation of tone

10.1 Representation of tones in (local) linguistic tradition

Present concisely the tradition of description of tone in the language (if any) and try to put the traditional notions and terminology in correspondence with those used in the current model.

This section is intended to cover both (a) vernacular terminology used by native linguists; and (b) traditional terminology specific to the study of the language/family/area, which may not be familiar to general linguists or experts on other tonal languages.

10.2 Tonal notation in the writing

Are tones marked in the practical orthography of the language in question (if any)? If they are, describe the principles of the practical tonal notation. How does it correspond to the “tonemic” notation?

If the tonal notation is “indirect” (the orthography is not designed to distinguish tones, but still gives some information about them), please, mention this fact.

If possible, provide the following information (very shortly):

- when the writing system was created;
- what is its real status in the language community (is it currently used in different spheres of life, or is it only known to a few activists of the literacy campaign, etc.).

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